SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	D.Sravanthi	**Reg. No.:**	192425315
Section / Batch:	2024-AI & ML	**Date:**	6.08.2026

Code of Conduct

I __D.Sravanthi(192425315)______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___D.Sravanthi_______________

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.

3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

1.Given

- (C(\text{data}) = 3)
- (C(\text{data science}) = 3)

Calculation

[P(\text{science}|\text{data})=\frac{3}{3}=1.0]

Answer

Bigram Probability = 1.0 (100%)

Interpretation

- The word **"science"** always follows **"data"** in the training corpus.
- Therefore, when a user types **"data"**, the keyboard confidently predicts **"science"** as the next word.
- A higher MLE probability results in **more accurate and faster next-word prediction**, improving typing speed and user experience.

2. Backoff Model for the unseen sequence "data science improves"

The trigram **"data science improves"** is **not found** in the corpus.

Step 1: Trigram

[P(\text{improves}|\text{data science}) = 0]

Since the trigram is unseen, backoff is applied.

Step 2: Bigram

The bigram

[P(\text{improves}|\text{science})]

is also unseen.

Step 3: Unigram

Backoff uses the unigram probability.

[P(\text{improves})]

Since its frequency is unavailable, only the unigram estimate is used.

Estimated Probability

[P(\text{data science improves})\approx P(\text{improves})]

Why is Backoff important?

- Handles unseen word combinations.
- Prevents prediction probability from becoming zero.
- Improves robustness of predictive keyboards.
- Allows meaningful suggestions even for new or rare phrases.
- Essential for real-time typing where users often enter unseen text.

3. Deleted Interpolation for "data science is"

Formula

[P=\lambda_1P_{tri}+\lambda_2P_{bi}+\lambda_3P_{uni}]

Given

- $\lambda_1=0.5$
- $\lambda_2=0.3$
- $\lambda_3=0.2$

Trigram Probability

$$P_{tri}=\frac{C(\text{science is})}{C(\text{data science})}
=\frac{2}{3}=0.667$$

Bigram Probability

$$P_{bi}=P(\text{is}|\text{science})
=\frac{2}{3}=0.667$$

Unigram Probability

From the given prediction statistics,

$$P_{uni}=P(\text{is})=0.66$$

Calculation

$$P=(0.5\times0.667)+(0.3\times0.667)+(0.2\times0.66)$$

$$=0.3335+0.2001+0.132$$

$$=0.6656$$

Final Answer

$$\boxed{P(\text{data science is})\approx0.666}$$

Discussion

Deleted Interpolation combines:

- Trigram information (highest accuracy)
- Bigram information (moderate reliability)
- Unigram information (maximum coverage)

This avoids zero probabilities and performs well even with sparse datasets.

4. Entropy Calculation

Formula

$$H=-\sum P(x)\log_2P(x)$$

Given

- $P(\text{is})=0.66$
- $P(\text{drives})=0.33$

Calculation

For "is"

$$-(0.66)\log_2(0.66)=0.395$$

For "drives"

$$-(0.33)\log_2(0.33)=0.528$$

Total Entropy

$$H=0.395+0.528$$

$$H=0.923\text{ bits}$$

Interpretation

- Entropy ≈ **0.92 bits**

- Low entropy indicates higher prediction confidence.
- Since **"is"** has the highest probability, the keyboard predicts it confidently.
- Entropy helps measure uncertainty; lower entropy leads to better predictive text performance.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

1. Penn Treebank POS Tags

Sentence 1

Book a flight ticket now.

Word	POS Tag	Meaning
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Sentence 2

This book is interesting.

Word	POS Tag	Meaning
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Justification

- In Sentence 1, **Book** is an action (command), so it is tagged **VB**.
- In Sentence 2, **book** refers to an object, so it is tagged **NN**.

2. HMM Probability

Formula

$$P(VB)=P(Start\rightarrow VB)\times P(book|VB)$$

Given

- $P(Start\rightarrow VB)=0.5$
- $P(book|VB)=0.6$

Calculation

$$P(VB)=0.5\times0.6$$

$$=0.30$$

Similarly,

$$P(NN)=0.5\times0.4=0.20$$

Answer

- Verb Probability = **0.30**
- Noun Probability = **0.20**

Since

$$0.30>0.20$$

the HMM assigns **VB (Verb)**.

Explanation

The probabilistic model favors **VB** because both the transition probability and emission probability together produce a higher likelihood.

3. Rule-Based vs HMM Tagging

Rule-Based Tagging	HMM Tagging
Uses handcrafted grammar rules	Uses probabilities from corpus
Difficult to maintain	Learns automatically
Performs poorly on ambiguous words	Resolves ambiguity effectively
Not scalable	Highly scalable
Less flexible	More adaptive

Recommendation

HMM Tagging is recommended because:

- Handles ambiguity better.
- Suitable for large datasets.
- More accurate.
- Learns from real language data.
- Ideal for modern chatbot deployment.

4. Importance of Standardized POS Tagsets

Standard POS tagsets:

- Provide consistent grammatical labeling.
- Improve intent detection.
- Help identify nouns, verbs, adjectives, and entities.
- Enhance sentence understanding.
- Improve chatbot response generation.
- Increase NLP accuracy.
- Reduce ambiguity.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

1. Apply the Transformation Rule

Initial Tags

Word	Initial Tag
economic	JJ
growth	NN
increases	NNS
employment	NN

Rule

Change NNS → VBZ if the previous word is NN.

Previous word before **increases** is **growth (NN)**.

Therefore,

NNS → VBZ

Updated Tags

Word	Final Tag
economic	JJ
growth	NN
increases	VBZ
employment	NN

Explanation

The sentence means

Economic growth increases employment.

Here **increases** is a **verb**, not a plural noun.

2. Analyze the Tagging Error

Initial Error:

- "increases" was tagged as **NNS (plural noun)**.

Correct Tag:

- **VBZ (third-person singular verb)**

Justification

Sentence structure:

Subject + Verb + Object

- Subject → Economic growth
- Verb → increases
- Object → employment

Thus VBZ is grammatically correct.

3. Word Frequency Distribution

Total Words

[120+450+210+380=1160]

Distribution

Word	Frequency	Probability
economic	120	120/1160 = **0.103**
growth	450	450/1160 = **0.388**
increases	210	210/1160 = **0.181**
employment	380	380/1160 = **0.328**

Explanation

Higher-frequency words receive stronger statistical support, allowing probabilistic POS taggers to assign more accurate tags based on observed corpus usage.

4. Transformation-Based Tagging and Entropy

Advantages of Transformation-Based Tagging

- Corrects errors after initial tagging.
- Learns linguistic rules automatically.
- Improves overall tagging accuracy.
- Handles contextual ambiguity.
- Produces more grammatically correct sentences.

Role of Entropy

Entropy measures uncertainty in assigning POS tags.

- **Before correction:** Higher entropy because "increases" could be either a noun or a verb.
- **After applying the transformation rule:** Lower entropy because the correct tag (VBZ) is confidently selected.

Lower entropy indicates **higher confidence** and **better tagging accuracy**, leading to improved news analytics and NLP performance.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________